Alexander Friedrichsen

Data Scientist & Applied AI Engineer

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ABOUT

Data scientist building production AI systems at the intersection of machine learning, automation, and software engineering. Drove AI adoption at a mutual insurance company — standing up production ML pipelines, agentic workflows, and a modern dev environment. Published ML researcher with experience shipping full-stack data products, ETL pipelines, and dashboards. Open to relocation (NYC, SF). Secret clearance.

WORK EXPERIENCE

Data Scientist and Analyst, Union Mutual
May, 2025 - Present
Data scientist and AI lead at a mutual insurance company. Co-built production ML systems and drove organization-wide AI adoption.

- Co-designed and deployed an AI-augmented inspection evaluation pipeline — computer vision model analyzes property and inspection images, rates risk, detects issues, and generates underwriter reports — reducing company losses by over $3M annually
- Drove AI adoption company-wide as sole internal AI advocate: introduced version control (GitHub), established production vs. test environments, restructured codebases for agent integration, and delivered hands-on agentic coding workshops to technical and non-technical staff
- Conduct exploratory modeling, feature engineering, and SQL querying across AWS (Redshift, S3) to support business decision-making and data accessibility
- Automate dashboards and monthly reports in Power BI and Python, delivering actionable KPI insights to cross-functional stakeholders

Software Engineer II, BAE Systems
Aug, 2023 - May, 2024 9 months

- Built multi-source Power BI data models with DAX aggregations, delivering weekly dashboards that drove labor and product line decisions for senior stakeholders
- Architected internal staffing solutions using SharePoint and MS Access, serving 16,000 employees
- Spearheaded ETL pipeline integrating financial data, enabling transition from manual Excel analysis to predictive analytics

Software Developer, Social Ecological Gaming and Simulation Lab
Aug, 2022 - Jul, 2023 11 months

- Developed precision agriculture simulation software using Python, C#, and JavaScript in Unity Game Engine

Public Health Analyst I, Vermont Department of Health
Feb, 2019 - Sep, 2021 2 years 7 months

- Automated reporting workflows (SAS, R) and published data products for CDC reporting systems, saving 10+ hours per month

PROJECTS EXPERIENCE

Multi-Agent AI Orchestration System
- Present
Built a production multi-agent system with 12 specialized AI agents coordinated through an Obsidian knowledge vault. Includes role-based task delegation, automated quality gates, persistent memory, MCP tool servers, and Discord-based always-on agent infrastructure.

- 12 specialized agents with model routing (judgment/execution/support tiers)
- Knowledge graph integration, automated vault search, session persistence
- Custom hooks for quality enforcement, secret detection, and agent verification

MTG Skill Analyzer
- Present
Full-stack web application analyzing 20,000+ competitive Magic: The Gathering match outcomes. Deployed at honestafblog.com with Supabase backend.

- ELO rating system, player performance modeling, data-driven team strategy

SKILLS

AI & Machine Learning

LLM Integration | Agentic Workflows | Computer Vision | Prompt Engineering | RAG | Structured Extraction | LLM Fine-Tuning | A/B Testing | Causal Inference | Statistical Modeling | Scikit-learn

Programming

Python | SQL | JavaScript | TypeScript | R | Julia | SAS

Data & Infrastructure

AWS (Redshift, S3) | Docker | Cloud Infrastructure | Power BI (DAX) | Google BigQuery | PostgreSQL | Supabase | ETL Pipelines

Development Practices

Git/GitHub | CI/CD | Full-Stack Development | Agile | Unit Testing | MCP (Model Context Protocol) | Agent Orchestration

EDUCATION

Data Science and Complex Systems, M.S., University of Vermont, Complex Systems Center
Aug, 2021 - May, 2023

Data Science, B.S., University of Vermont, College of Engineering and Mathematics
Aug, 2018 - May, 2022

Minor: Economics | Minor: Mathematics | Minor: Computer Science | Minor: Statistics

PUBLICATIONS

Evolutionary Machine Learning for Robust Facility Placements, GECCO '23 (Genetic and Evolutionary Computation Conference)
Published on: Jul 01, 2023
Published research on evolutionary algorithms for optimization under uncertainty. Also presented at JuliaCon.

INTERESTS

Competitive Card Games

Pro Tour Magic: The Gathering competitor ($500K events) | Team Captain — Team Handshake | World Champion — Wonders of the First